CO_2 SPAR RECHNER

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ABOUT THE APP

The Co2 Spar Rechner is the backend for the **YoungCaritas Kleidertausch CO_2 Savings Calculator**, developed as a final semester project to improve the original system with:

- A complete admin dashboard
- User registration & approval
- Persistent data saving via PostgreSQL
- A flexible CO_2 item management system

WHY?

Previously, the YoungCaritas CO_2 calculator had issues like:

- Too simple with limited item options
- No user management
- No data persistence

The concept of "saving CO_2 through reuse" is socially impactful and fits YoungCaritas' mission and I offered to make it actually usable by during Kleidertausch events.

WHAT I DID

- Rebuilt the backend using **FastAPI + PostgreSQL**

Added:

- An Admin dashboard
- Secures user registration/approval
- CO_2 item database (editable)
- Swagger API and UI views
- Made it scalable and shareable between locations

DEVELOPMENT PROCESS

- **Planning:** Defined what features the original system lacked
- **Design:** Set up database models, authentication flow, admin views
- **Implementation:** Built API endpoints, templates, and email system
- **Testing:** Used pytest and Swagger to validate functionality
- **Deployment-Ready:** Dockerized the app for team or production use

RESULTS & IMPACT

- Fully working CO_2 tracking backend
- Live admin UI: easy to add/edit clothing items
- Local event teams can use and manage their own calculator
- Tangible impact: encourages reuse, quantifies environmental benefit
- Ready to be deployed across multiple YoungCaritas chapters
- Connects with the frontend I built to display the items from the dtatabase.

CONCLUSIONS & FUTURE WORK

- App works well and solves the key problems from the original system.

Future Improvements:

- Add statistics/dashboard for total saved CO_2
- Add multi-language support (e.g. English/German)
- Allow users to suggest items to admins
- Host on cloud (e.g., Render, Heroku)
- Real-world use: Can be handed over to YoungCaritas to use across events

THANK YOU